Introduction to mathematics for economists

Exercise Sheet #4 — Mathcamp 2026, Sciences Po Master's in Economics

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2026-09-03

During the tutorials, we will solve exercices 1, 2, 5 and 6. Prioritize these. The unconstrained optimization warm-up is a recommended first step before tackling the Lagrangian and Kuhn–Tucker exercises. A full answer sheet is provided, since we will not have time to solve everything in class.

Two results are used repeatedly across the sheet.

Weierstrass (extreme value theorem)

Let $K \subset \mathbb{R}^n$ be **nonempty** and **compact** (i.e. closed and bounded), and let $f : K \to \mathbb{R}$ be continuous. Then f attains both a maximum and a minimum on K.

Sufficiency of the first-order conditions

If f is **concave** and the feasible set is **convex**, then any point satisfying the first-order (Lagrange or Kuhn–Tucker) conditions is a *global* maximum. If in addition f is **strictly** concave, the maximiser is unique.

Exercise 1: the Budget Set

Consumption Set and Budget Set

Let $X = \mathbb{R}^n_+$ be the **consumption set** of a price-taking economic agent, and let

$$B(p, w) = \{x \in \mathbb{R}^n_+ \mid \langle p, x\rangle \leq w\}$$

be his **budget set**, where $p = (p_1, \dots, p_n) \in \mathbb{R}^n_{++}$ is a price vector and $w > 0$ is the wealth of the consumer. Recall that $\langle p, x\rangle$ stands for the inner product of p and x in $\mathbb{R}^n$, i.e. $\langle p, x\rangle = \sum_{i=1}^n p_i x_i$.

1. Prove that, for any $\lambda > 0$, we have $B(\lambda p, \lambda w) = B(p, w)$.
2. Prove that $B(p, w)$ is a convex set, i.e. for any $x, y \in B(p, w)$ and for any $\mu \in [0, 1]$, $z = \mu x + (1 - \mu)y \in B(p, w)$.
3. Prove that $B(p, w)$ is a compact set (i.e. bounded and closed). Remember that a subset A of $\mathbb{R}^n_+$ is bounded if there exists $M > 0$ such that for any $x \in A$ we have $\|x\| \leq M$, where $\|\cdot\|$ is the Euclidean norm, $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$.

4. Using questions 2 and 3, explain why the following constrained optimization program always has a solution:

$$\begin{cases} \max\limits_{x \in \mathbb{R}^n_+} u(x) \\ \text{s.t. } \langle p, x \rangle \leq w \end{cases}$$

where $u : \mathbb{R}^n_+ \to \mathbb{R}$ is a continuous utility function.

5. Suppose there are two goods x_1, x_2 with prices p_1, p_2 and budget w. Assume the economic agent has utility

$$U(x_1, x_2) = \ln(x_1^\alpha x_2^\beta), \qquad \alpha > 0, \ \beta > 0,$$

and wants to maximize it subject to $x_1 p_1 + x_2 p_2 = w$.

a. Is this optimization problem parametric or not?
b. Express x_2 as a function of x_1.
c. Solve the equivalent unconstrained optimization problem.

Exercise 2: OLS

The Least-Squares Problem

We observe $n \geq 2$ pairs $(x_1, y_1), \dots, (x_n, y_n)$ of real numbers and wish to summarise them by a straight line $y \approx a + bx$. For a given intercept a and slope b, the **residual** of observation i is $e_i = y_i - a - bx_i$, and the fit is measured by the sum of squared residuals

$$S(a, b) = \sum_{i=1}^{n} (y_i - a - bx_i)^2.$$

The **ordinary least squares** (OLS) estimator is $\min\limits_{(a,b) \in \mathbb{R}^2} S(a, b)$.

Note what is being optimised over: the choice variables are the *parameters* (a, b), while the data (x_i, y_i) are fixed. There is no constraint: (a, b) ranges over all of $\mathbb{R}^2$.

Write $\bar{x} = \frac{1}{n} \sum_i x_i$ and $\bar{y} = \frac{1}{n} \sum_i y_i$.

1. Compute the gradient of S and write the first-order conditions. Show that the first one says the fitted line passes through $(\bar{x}, \bar{y})$.
2. Compute the Hessian of S. Show that it does not depend on (a, b), and that it is positive definite **if and only if** the x_i are not all equal.
3. Assume the x_i are not all equal. Solve the first-order conditions, and explain why the solution is the unique **global** minimum.
4. Suppose instead that all the x_i are equal to some common value c. Describe the set of minimisers of S. Interpret.

Exercise 3: equality constraints

Lagrangian with Equality Constraints

To maximize (or minimize) $f(x)$ subject to equality constraints $g_1(x) = 0, \dots, g_m(x) = 0$, form the **Lagrangian**

$$L(x, \lambda) = f(x) - \sum_{j=1}^{m} \lambda_j \, g_j(x)$$

and look for points satisfying $\nabla_x L = 0$ together with the constraints, provided the constraint gradients $\nabla g_1(x), \dots, \nabla g_m(x)$ are linearly independent at the candidate point (*constraint qualification*). The nature of each critical point (max, min, or saddle) is then assessed using the **bordered Hessian**.

Bordered Hessian sign rule

With n choice variables and m constraints, form the $(n + m) \times (n + m)$ bordered Hessian

$$H_b = \begin{bmatrix} 0_{m \times m} & Dg \\ (Dg)^\top & D^2_{xx} L \end{bmatrix}, \qquad Dg = \begin{bmatrix} \nabla g_1^\top \\ \vdots \\ \nabla g_m^\top \end{bmatrix}.$$

One checks the last $n - m$ leading principal minors. The candidate is a local **maximum** if they alternate in sign starting with the sign of $(-1)^n$, and a local **minimum** if they all have the sign of $(-1)^m$.

We want to find the point of the circle of $\mathbb{R}^3$ defined by

$$x^2 + y^2 + z^2 = 1, \qquad x + y + z = 0$$

whose distance to $(1, 1, 0)$ is maximal, and the point whose distance is minimal.

1. Write the Lagrangian of the problem. *(Hint: you need to define the objective first... a careful choice should avoid square roots)*
2. Compute the critical points, and check that at these points the constraints are qualified.
3. Determine which is the maximum and which is the minimum.

Exercise 4: inequality constraints

Kuhn–Tucker Conditions

For $\max_x f(x)$ subject to $h_k(x) \leq 0$ for $k = 1, \dots, K$, form

$$L(x, \lambda) = f(x) - \sum_{k=1}^{K} \lambda_k \, h_k(x),$$

The Kuhn–Tucker (KKT) conditions require, at a candidate optimum $x^\star$:

- *stationarity*: $\nabla_x L(x^\star, \lambda) = 0$;
- *primal feasibility*: $h_k(x^\star) \leq 0$ for every k;
- *dual feasibility*: $\lambda_k \geq 0$ for every k;
- *complementary slackness*: $\lambda_k \, h_k(x^\star) = 0$ for every k.

These conditions are **necessary** at an optimum provided a constraint qualification holds (e.g. the gradients of the *active* constraints are linearly independent). When in addition f is concave and the feasible set is convex, they are also **sufficient** for a global maximum.

We consider the problem

$$\min_{x \geq 0,\, y \geq 0,\, x+y \geq 4} (3x^2 + 4xy + 5y^2).$$

1. Show that it admits at least one solution.
2. Rewrite it as the maximization problem

$$\max_{x \geq 0,\, y \geq 0,\, x+y \geq 4} -(3x^2 + 4xy + 5y^2)$$

   and write the Kuhn–Tucker conditions.
3. Show that the constraint $x + y \geq 4$ is binding at the optimum.
4. Check the constraint qualification.
5. Find the unique solution.

Exercise 5: application with Cobb-Douglas utility

A rational, price-taking consumer has preferences represented by a **Cobb–Douglas utility function**:

$$u(x) = \prod_{i=1}^{n} x_i^{\alpha_i}, \quad \text{for all } x \in \mathbb{R}^n_+, \text{ where } \alpha_i > 0 \text{ for every } i \text{ and } \sum_{i=1}^{n} \alpha_i < 1.$$

Market prices are $p = (p_1, \dots, p_n) \in \mathbb{R}^n_{++}$ and the wealth of the consumer is $w > 0$.

1. Justify why the budget constraint must be binding at the optimum.
2. Justify why the optimal solution must be interior, i.e. all quantities must be strictly positive at the optimal bundle $x^\star$.
3. Justify why the optimal solution of the consumer's problem must be unique.
4. Use the Kuhn–Tucker method to solve the consumer's problem and characterize the Marshallian demands $x_i(p, w)$.

We now assume $X = \mathbb{R}^2_+$, so bundles $x = (x_1, x_2)$ and price vectors $p = (p_1, p_2)$ are two-dimensional.

5. Represent approximately the shape of the budget set $B(p, w)$.
6. Assume the State raises a lump-sum tax T, with $0 < T < w$, on the wealth of the consumer. Represent graphically the effect of this tax on the budget set.
7. Suppose instead that the State raises a VAT at rate $t > 0$ on the consumption of good 2. Represent the effect of such a tax on the budget set compared to the initial situation with no tax.
8. How does each tax change the optimal solution? Which of the two would you expect to hurt the consumer more, for a given amount of revenue raised?

Exercise 6: application with Stone–Geary utility

Stone–Geary Preferences

A rational, price-taking consumer has preferences represented by

$$\forall (x_1, x_2) \in \mathbb{R}^2_+, \quad u(x_1, x_2) = \ln(x_1 + \gamma_1) + \ln(x_2 + \gamma_2),$$

where $\gamma_1, \gamma_2 > 0$ are given parameters. Market prices are $p = (p_1, p_2) \in \mathbb{R}^2_{++}$ and wealth is $w > 0$.

Interpret γ_i as a **free endowment**: the consumer behaves as though he already owned γ_i units of good i before going to the market, and only cares about his *total* holding $x_i + \gamma_i$. Because utility is already finite and well defined at $x_i = 0$, neither good is essential here.[a]

[a] The textbook Stone–Geary / Linear Expenditure System uses $\sum_i \beta_i \ln(x_i - \gamma_i)$ with a *minus* sign, where γ_i is a *subsistence minimum*, goods are essential and corner solutions are impossible. The sign flip is exactly what makes this exercise interesting.

1. Represent graphically some indifference curves of this utility function. Is it possible that the consumer's problem has corner solutions?
2. Use the Kuhn–Tucker method to solve the consumer's problem and characterize the Marshallian demands.
3. Prove that the Marshallian demand functions are homogeneous of degree zero.

Homogeneous Function

A function $f : \mathbb{R}^n \to \mathbb{R}$ is **homogeneous of degree** $k \in \mathbb{R}$ if, for all $\lambda > 0$ and all $x \in \mathbb{R}^n$,

$$f(\lambda x) = \lambda^k f(x).$$